

BABU BANARASI UNIVERSITY

Descriptive analytics project file



BLINKIT DATASET

GROUP-01

ROLL NOs

MOHD ARSALAN

1240258262

MOHD ZAID KHAN

1240258275

MOHD TAHA KHAN

1240258273

MO ASIF

1240258260

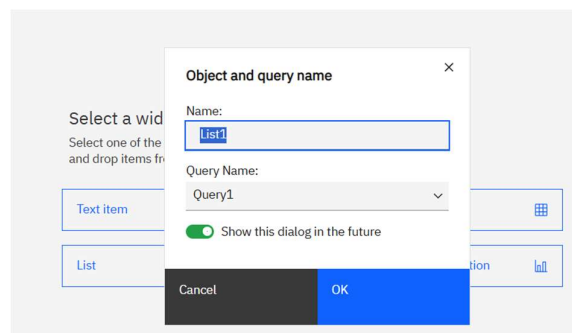
PROJECT

Problem Statement 1 :- Total Order Value Per Order : Calculate the total value of each order by multiplying `quantity * unit_price` and grouping by `order_id`

Solution :-

Step 1 :- Open Report

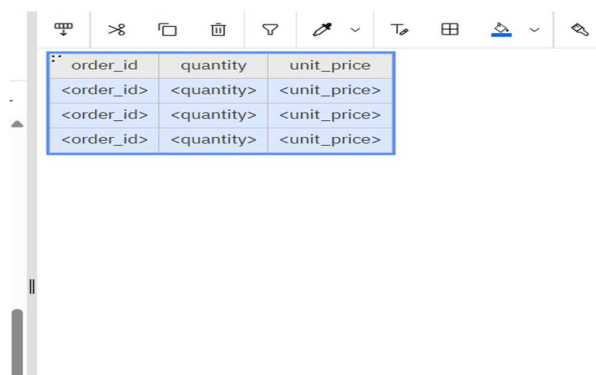
Create or open a **List Report** in Cognos



Step 2 :- Add Data Items

Drag **order_id** into the list.

Drag **quantity** and **unit_price** into the list as well

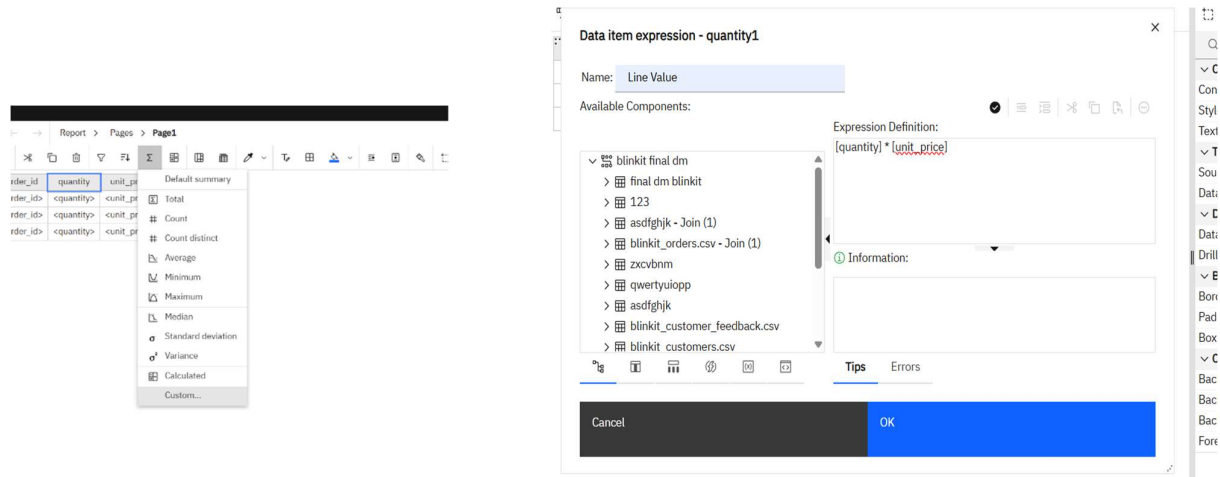


order_id	quantity	unit_price
<order_id>	<quantity>	<unit_price>
<order_id>	<quantity>	<unit_price>
<order_id>	<quantity>	<unit_price>

Group No. 01

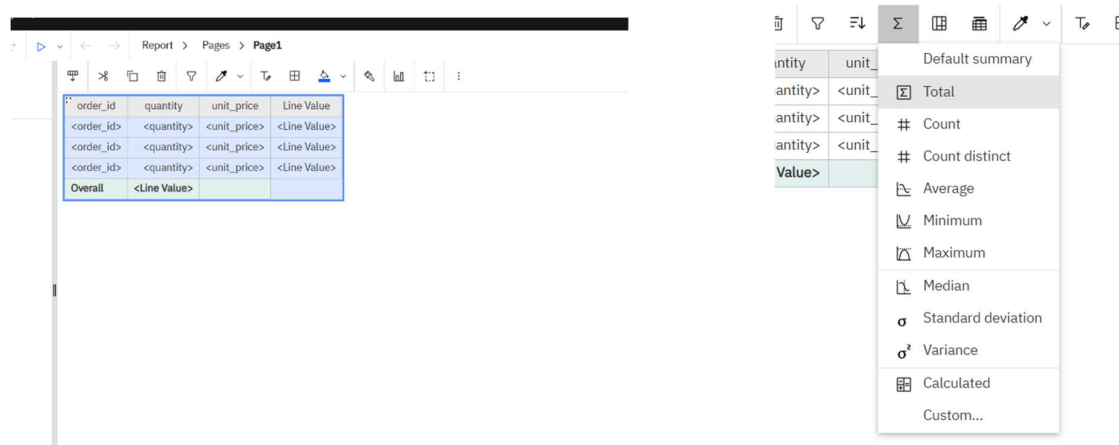
Step 3 :- Create a Calculation for Line Value

- From the toolbar, go to **Data** → **Calculation**.
- In the expression editor, enter:
- Name it **Line_Value** (this represents the value of each item row in the order).



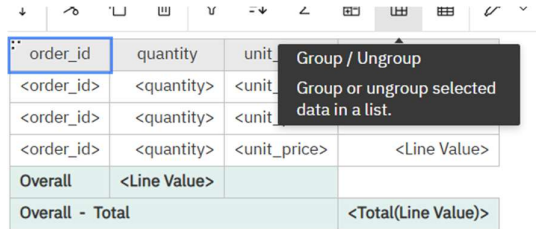
Step 4 :- Aggregate to Get Order Total

- Drag **Line_Value** into the list.
- Click on it → choose **Aggregate Function** → **Total**.



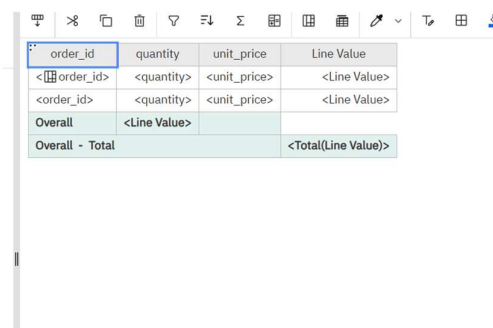
Step 5 :- Group by Order_ID

- Click on the **order_id** column → choose **Group** from the toolbar.
- This ensures each order is summarized.



The screenshot shows a data table with columns: order_id, quantity, unit_price, and Line Value. The 'order_id' column is selected, and a context menu is open with the options 'Group / Ungroup' and 'Group or ungroup selected data in a list.' The table structure is as follows:

order_id	quantity	unit_price	Line Value
<order_id>	<quantity>	<unit_price>	<Line Value>
<order_id>	<quantity>	<unit_price>	<Line Value>
<order_id>	<quantity>	<unit_price>	<Line Value>
Overall	<Line Value>		
Overall - Total			<Total(Line Value)>



The screenshot shows the result of grouping by 'order_id'. The table structure is as follows:

order_id	quantity	unit_price	Line Value
<order_id>	<quantity>	<unit_price>	<Line Value>
<order_id>	<quantity>	<unit_price>	<Line Value>
Overall	<Line Value>		
Overall - Total			<Total(Line Value)>

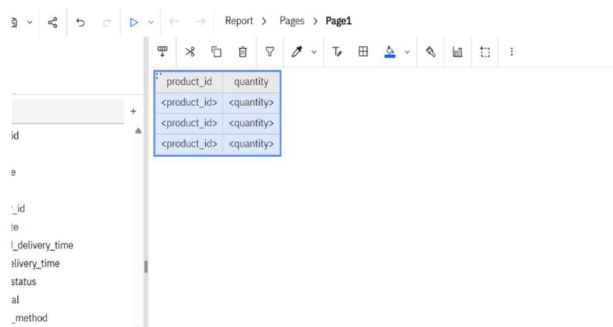
PROJECT

Problem Statement 2 :- Top-Selling Product by Quantity: identify the product with the highest total quantity ordered across all order using product_id and quantity

Solution :-

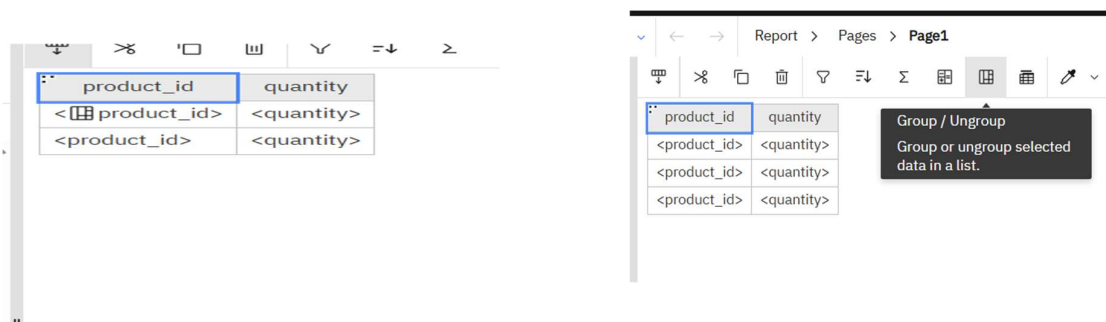
Step 1 :- Add Required Data Items

- Drag **product_id** into the list.
- Drag **quantity** into the list.



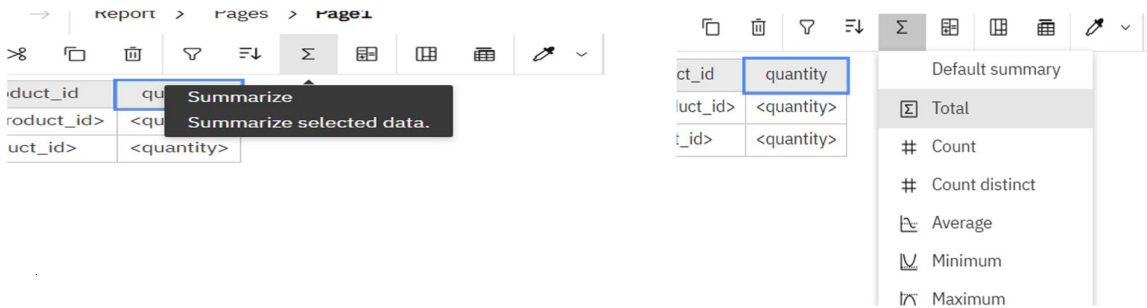
Step 2 :- Group by Product

- Click on **product_id** column → Right-click → **Group**.
- This will consolidate rows per product.



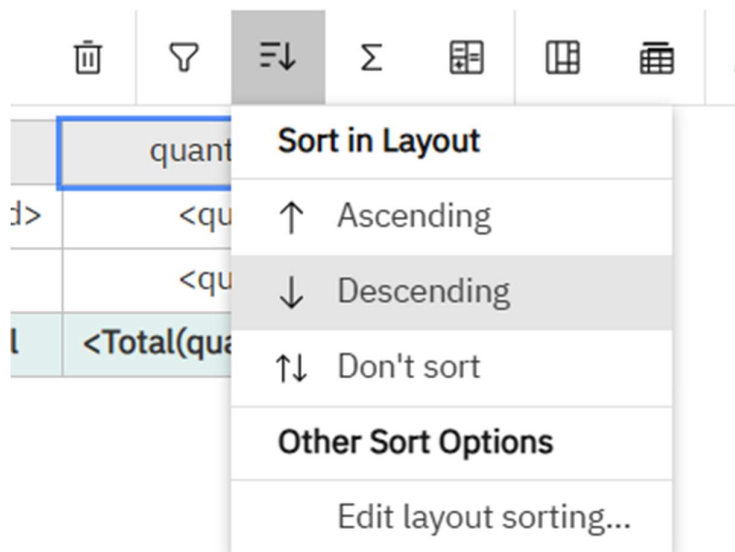
Step 3 :- Summarize Quantities

- Click on **quantity** column → **Aggregate** → **Total (Sum)**.
- Now you'll have **total quantity ordered per product_id**.



Step 4 :- Sort the Results

- Click on the **Total Quantity** column → choose **Sort** → **Descending**.
- This ensures the product with the highest order quantity comes first



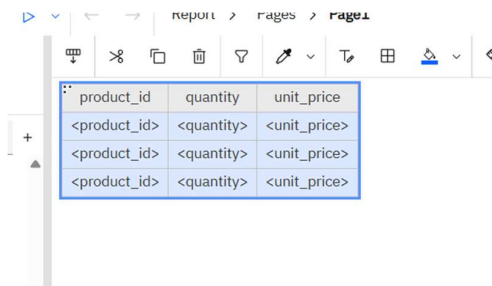
PROJECT

Problem Statement 3 :- Revenue Contribution per Product: Find the revenue contribution of each product by summing quantity * unit_price for each product_id.

Solution :-

Step 1 :- Add Data Items

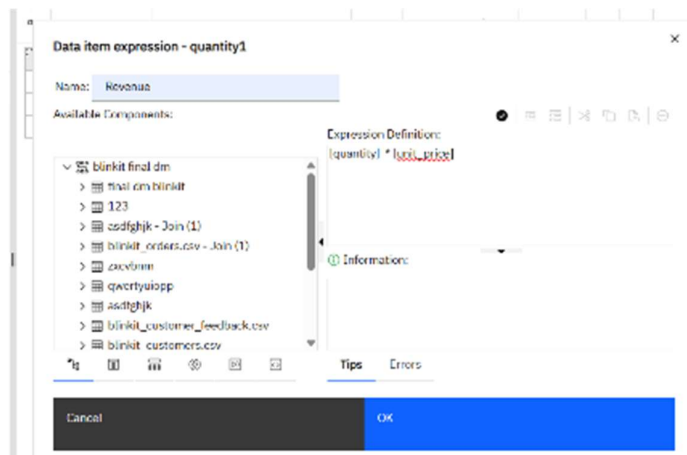
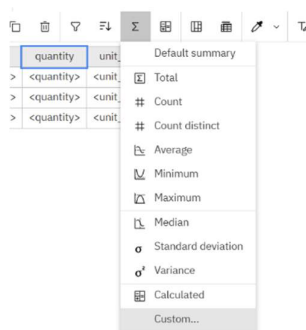
- Drag **product_id** into the list.
- Drag **quantity** and **unit_price**



product_id	quantity	unit_price
<product_id>	<quantity>	<unit_price>
<product_id>	<quantity>	<unit_price>
<product_id>	<quantity>	<unit_price>

Step 2 :- Create Revenue Calculation

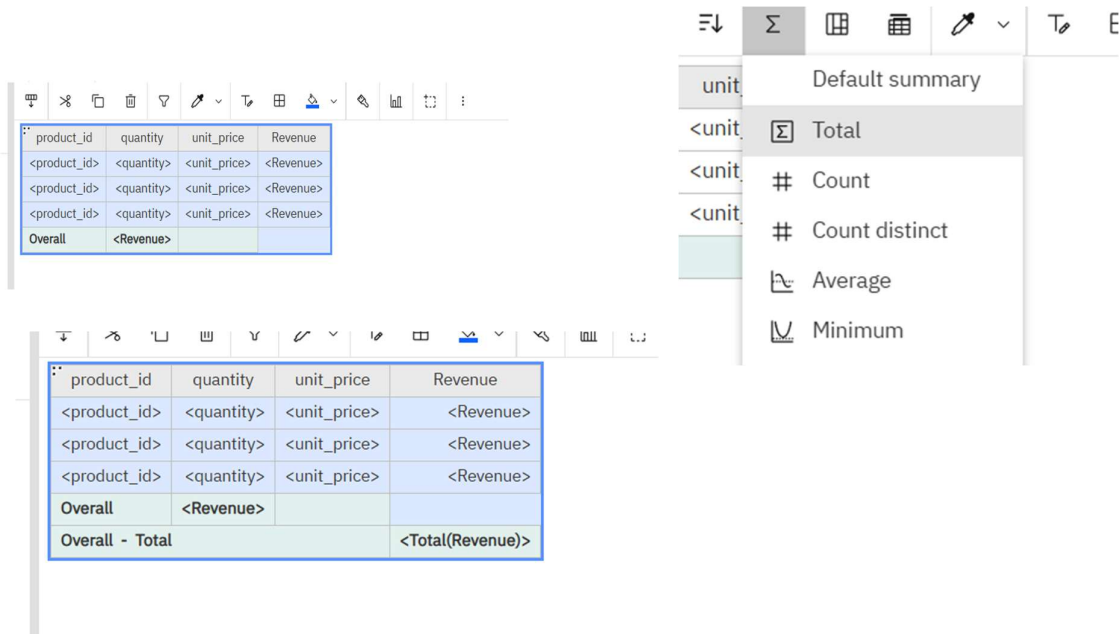
- From the toolbar: **Data** → **Calculation**.
- Name it **Revenue**.



Group No. 01

Step 3 :- Aggregate the Revenue

- Click on the **Revenue** column.
- From the toolbar: **Aggregate** → **Total (Sum)**



The screenshot shows a data table with four columns: product_id, quantity, unit_price, and Revenue. The table has five rows, including an 'Overall' summary row. A context menu is open over the 'Revenue' column, displaying various aggregation options. The 'Total' option is highlighted.

product_id	quantity	unit_price	Revenue
<product_id>	<quantity>	<unit_price>	<Revenue>
<product_id>	<quantity>	<unit_price>	<Revenue>
<product_id>	<quantity>	<unit_price>	<Revenue>
Overall	<Revenue>		

product_id	quantity	unit_price	Revenue
<product_id>	<quantity>	<unit_price>	<Revenue>
<product_id>	<quantity>	<unit_price>	<Revenue>
<product_id>	<quantity>	<unit_price>	<Revenue>
Overall	<Revenue>		
Overall - Total			<Total(Revenue)>

Context Menu Options:

- unit
- <unit>
- <unit>
- <unit>
- Default summary
- Total
- Count
- Count distinct
- Average
- Minimum

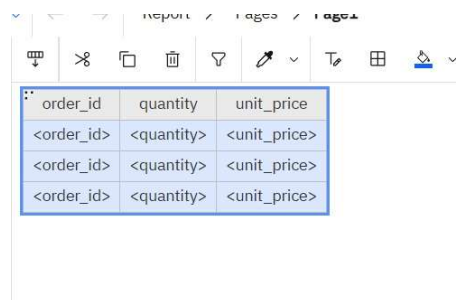
PROJECT

Problem Statement 4 :- Average Order Value: Compute the average order value for all order in the dataset.

Solution :-

Step 1 :- Add Data Items

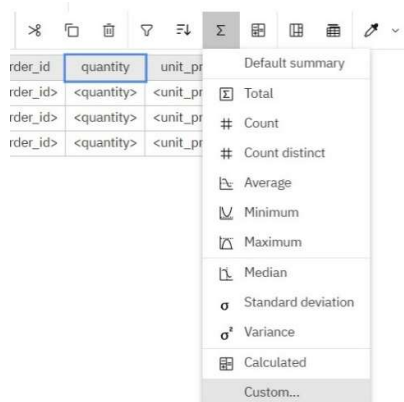
- Drag in order_id, quantity, and unit_price.



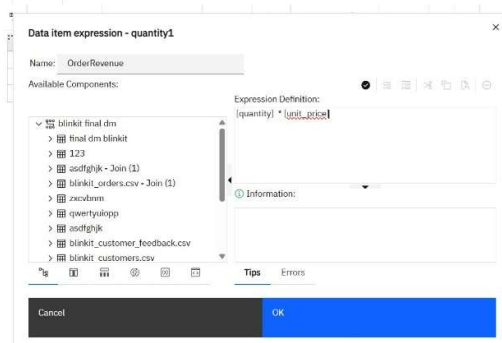
order_id	quantity	unit_price
<order_id>	<quantity>	<unit_price>
<order_id>	<quantity>	<unit_price>
<order_id>	<quantity>	<unit_price>

Step 2 :- Calculate Order Revenue

- Create a new calculation → name it OrderRevenue.
- Aggregate it → Total (Sum) grouped by order_id

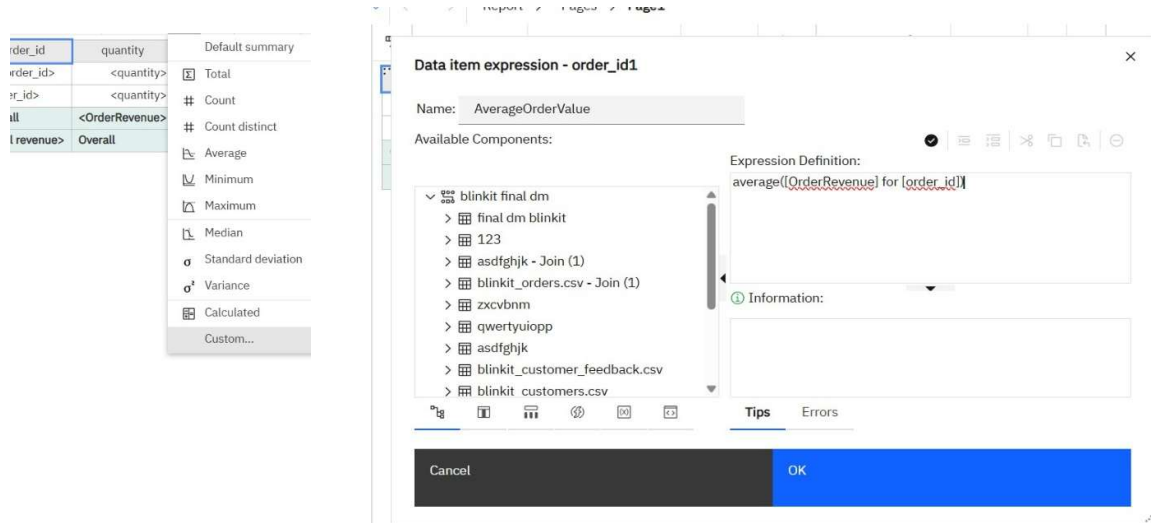


order_id	quantity	unit_price
<order_id>	<quantity>	<unit_price>
<order_id>	<quantity>	<unit_price>
<order_id>	<quantity>	<unit_price>



Step 3 :- Calculate Average Order Value

- Create another calculation → name it AverageOrderValue.
- This gives the average of all order totals.



PROJECT

Problem Statement 5 :- Order Size Distribution: Categorize order by number of item (quantity sum per order) - e.g. small,medium,large and count how many order fall into each category.

Solution :-

Step 1 :- Add Data Items

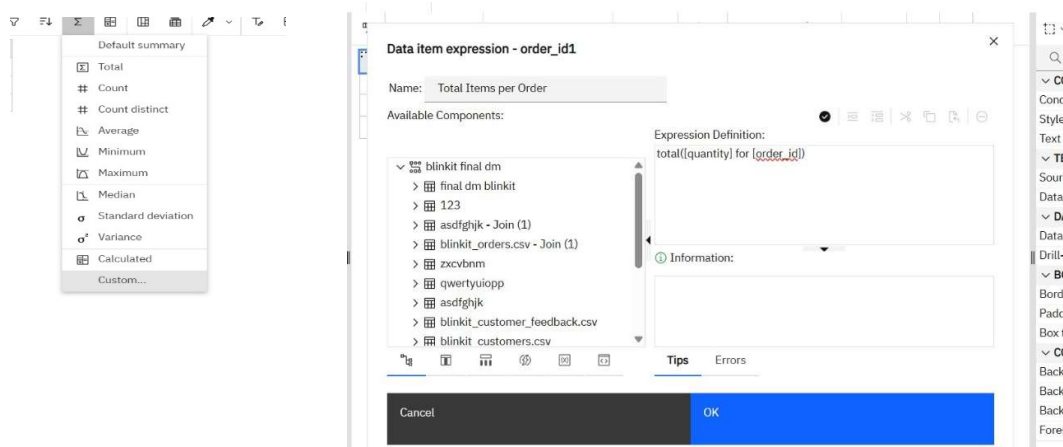
- Drag order_id and quantity

order_id	quantity
<order_id>	<quantity>
<order_id>	<quantity>
<order_id>	<quantity>

Step 2 :- Calculate Total Item per Order

- Create a new calculation → name it TotalItems
- Expression:

Total([quantity] for [order_id])



- This give sum of quantities per order

Group No. 1

Step 3 :- Create order Category

- Create another calculation → name it OrderCategory

Data item expression - order_id1

Name: OrderCategory.

Available Components:

- blinkit final dm
 - final dm blinkit
 - 123
 - asdfghjk - Join (1)
 - blinkit_orders.csv - Join (1)
 - zxcvbnm
 - qwertyuiopp
 - asdfghjk
 - blinkit_customer_feedback.csv
 - blinkit_customers.csv

Expression Definition:

```
case
when ([TotalItems] <= 10) then 'Small'
when ([TotalItems] between 11 and 50) then 'Medium'
else 'Large'
end
```

Information:

case ... end
Indicates the end of a case or when construct.

Cancel OK

Step 4 :- Count Order per Category

- Drag OrderCategory into the list

Query1

- order_id
- quantity
- Total Items per Order
- OrderCategory.

order_id	quantity	OrderCategory.
<order_id>	<quantity>	<OrderCategory.>
<order_id>	<quantity>	<OrderCategory.>
<order_id>	<quantity>	<OrderCategory.>
<Total Items per Order>	Overall	
<OrderCategory.>	Overall	

- Apply Aggregate → Count on Order_id

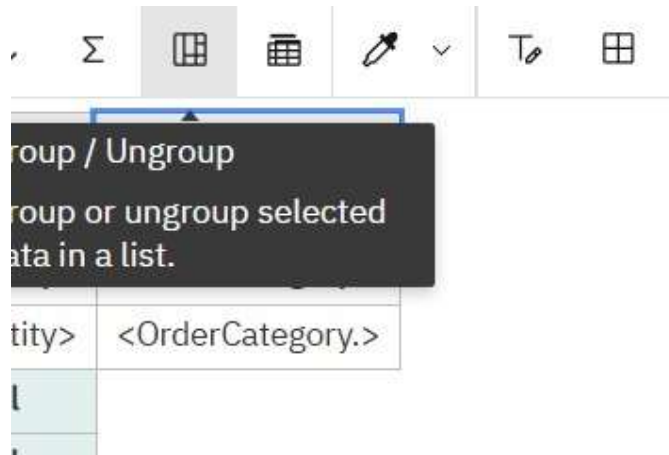
Report > Pages > Page1

Default summary

- Total
- Count
- Count distinct
- Average
- Minimum

order_id	quantity	
<order_id>	<quantity>	
<order_id>	<quantity>	
<order_id>	<quantity>	
<Total Items per Order>	Overall	
<OrderCategory.>	Overall	

- Group by OrderCategory



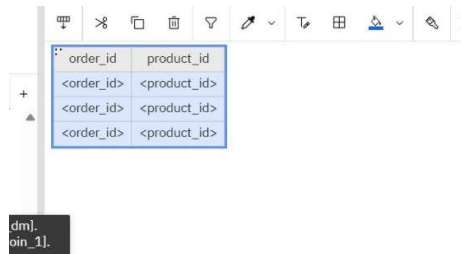
PROJECT

Problem Statement 6 :- Most Frequent Product Per Order: Find which product_id appears most frequently across order.

Solution :-

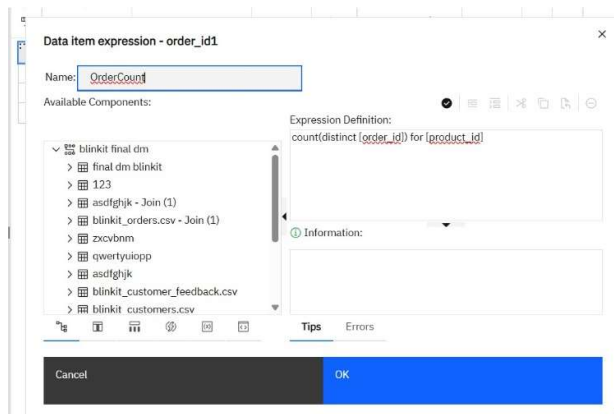
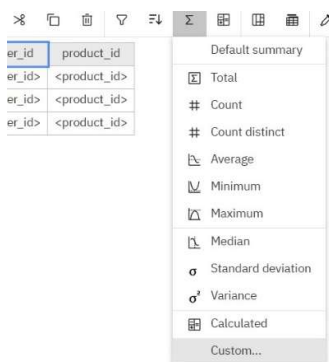
Step 1 :- Add Data Items

- Drag order_id and product_id.



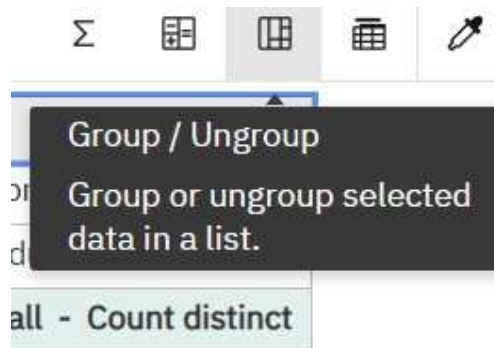
Step 2 :- Count Frequency of Each Product

- Create a calculation → name it OrderCount



Step 2 :- Group By Product

- Group the list by product_id.

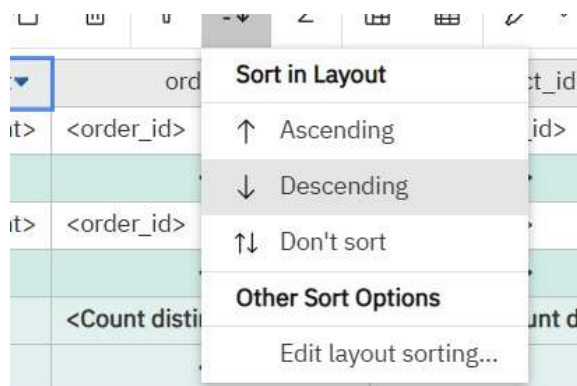


- Add **OrderCount** column

OrderCount	order_id	product_id
<OrderCount>	<order_id>	<product_id>
	<OrderCount>	<product_id>
<OrderCount>	<order_id>	<product_id>
	<OrderCount>	<product_id>
	<Count distinct(order_id)>	Overall - Count distinct
	<OrderCount>	Overall

Step 3 :- Sort Descending

- Sort **OrderCount** in descending order so the most frequent product appear first.



PROJECT

Problem Statement 7 :- High-Value Orders: identify order whose total order value exceeds a given threshold(e.g. rs5000)

Solution :-

Step 1 :- Add Data Items

- Drag order_id, quantity, and unit_price.

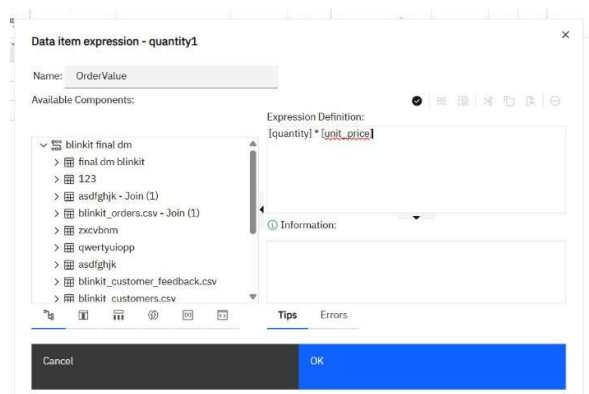
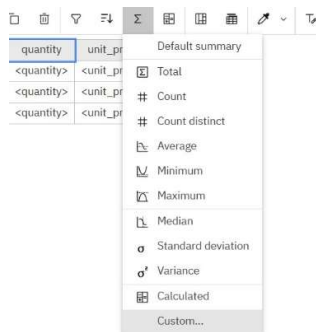


order_id	quantity	unit_price
<order_id>	<quantity>	<unit_price>
<order_id>	<quantity>	<unit_price>
<order_id>	<quantity>	<unit_price>

■

Step 2 :- Create order value Calculation

- Go to data → Calculation → Name it OrderValue



Group No.1

Step 3 :- Aggregate Order Value per Order

- Group the list by order_id.

order_id	quantity	unit_price
order_id>	<quantity>	<unit_price>
order_id>	<quantity>	<unit_price>
order_id>	<quantity>	<unit_price>
verall	<OrderValue>	

- On The OrderValue column → apply Aggregate → Total(sum)

OrderValue	order_id	quantity	unit_price
<OrderValue>	<order_id>	<quantity>	<unit_price>
<OrderValue>	<order_id>	<quantity>	<unit_price>
	Overall	<OrderValue>	

value	order_id	
value>	<order_id>	
value>	<order_id>	
	Overall	<C

- Now We have total order value per order_id

Query1
order_id
quantity
unit_price
OrderValue
Total(OrderValue)

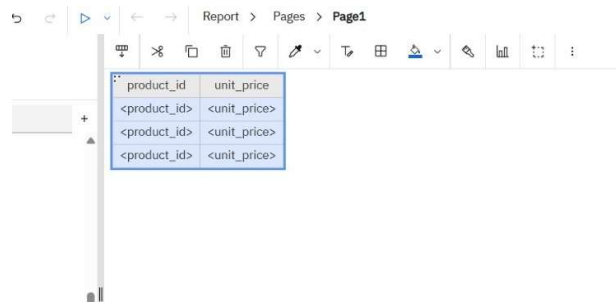
PROJECT

Problem Statement 8 :-Price Variation Across Product: Check which product have highest average unit_price and which have the lowest.

Solution :-

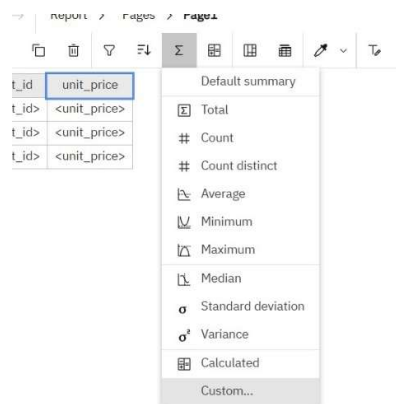
Step 1 :- Create the Base Report

- Open a List report.
- Drag product_id (or product_name) and unit_price into the list.

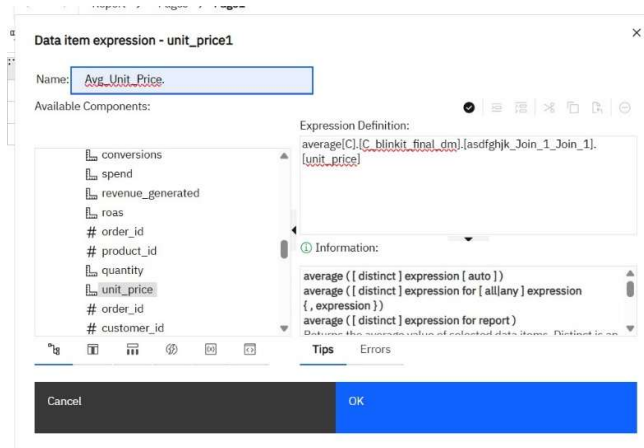


Step 2 :- Calculate the Average unit_price

- From the toolbar, go to Data → Calculation.
- In the expression editor, create a new calculation:

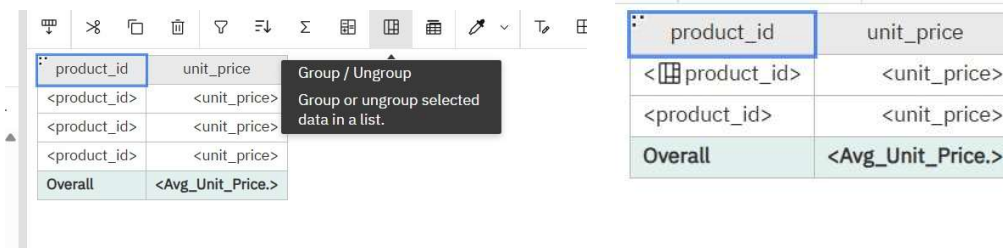


- Name it something like Avg_Unit_Price.
- Drag this new calculation into your list.



Step 3:- Group by Product

- Select the product_id column.
- Right-click → Group.



Step 4:- Find Highest and Lowest

- Click the Avg_Unit_Price column.
- Apply Sort Descending → the first row will be the highest average unit price.
- Apply Sort Ascending → the first row will be the lowest average unit price.

Group No. 1

product_id	Avg_Unit_Price.	unit_price
<product_id>	<Avg_Unit_Price.>	<unit_price>
<product_id>	<Avg_Unit_Price.>	<unit_price>
Overall		<Avg_Unit_Price.>

The screenshot shows the 'Sort in Layout' context menu in Tableau. The menu is open over the 'Avg_Unit' column header. The options visible are:

- Ascending (indicated by an upward arrow icon)
- Descending (indicated by a downward arrow icon)
- Don't sort (indicated by a double-headed arrow icon)
- Other Sort Options (a section header)
- Edit layout sorting... (an option under 'Other Sort Options')



The screenshot shows the 'Sort in Layout' menu in Tableau. The menu is open, displaying the following options:

- Ascending
- Descending
- Don't sort
- Other Sort Options**
 - Edit layout sorting...

The background shows a table with columns 'Avg_Unit' and 'Avg_Unit'.

PROJECT

Problem Statement 9 :-Quantity vs. Price Correlation: Check if there's any pattern between quantity purchased and unit_price (e.g. bulk discount or higher price for small quantities)

Solution :-

Step 1 :- Add the Required Columns

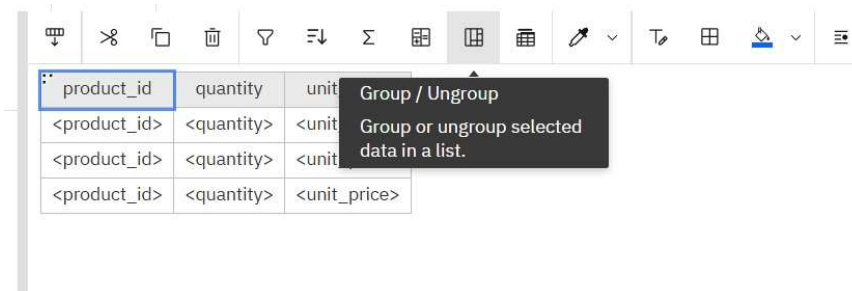
- In your List report, drag the following data items:
- Product_ID (or Product_Name)
- Quantity
- Unit_Price



product_id	quantity	unit_price
<product_id>	<quantity>	<unit_price>
<product_id>	<quantity>	<unit_price>
<product_id>	<quantity>	<unit_price>

Step 2:- Create Groupings

- Select the product_id → right click → Group



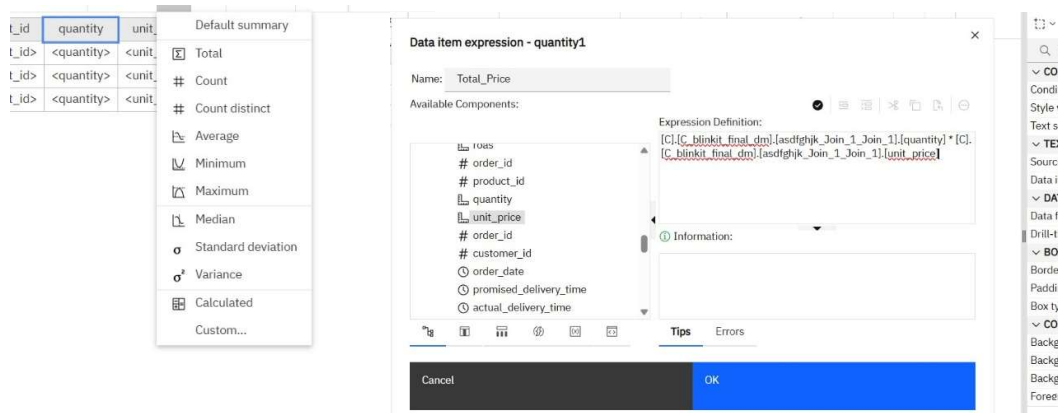
product_id	quantity	unit_price
<product_id>	<quantity>	<unit_price>
<product_id>	<quantity>	<unit_price>
<product_id>	<quantity>	<unit_price>

Group / Ungroup
Group or ungroup selected data in a list.

Group No.1

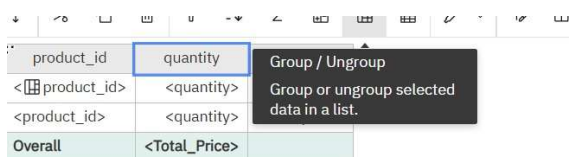
Step 3:- Create Calculations

- Average Unit Price per Quantit
- Name it Total_Price



Step 4:- Group by Quantity

- Select the Quantity column → right-click → Group → Group by this column.
- Add your new calculation Total_Price By_Quantity beside it.

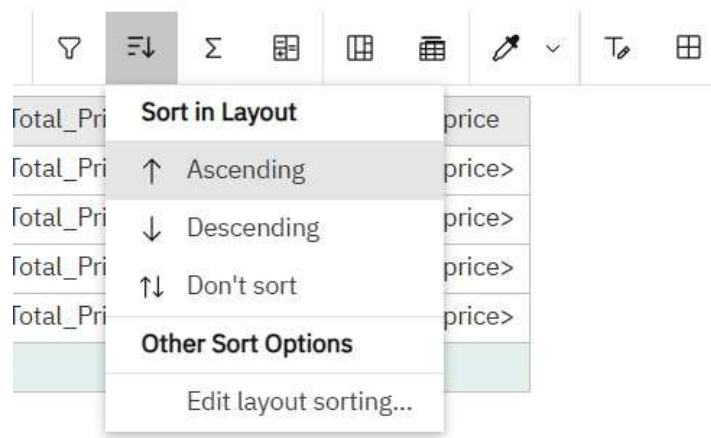


product_id	Total_Price	quantity	unit_price
<product_id>	<Total_Price>	<quantity>	<unit_price>
	<Total_Price>	<quantity>	<unit_price>
<product_id>	<Total_Price>	<quantity>	<unit_price>
	<Total_Price>	<quantity>	<unit_price>
Overall		<Total_Price>	

Group No. 1

Step 5:- Sort & Observe

- Sort the Quantity column in ascending order.

**Group No. 1**